LESSON PLAN

ARMAMENT STORE PROCEDURE (33)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	8	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Brief history and development of requirements of ejection seat
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Brief history and development of requirements of ejection seat
5. Specific Objectives	33.1.1.1 Trainee will be able to explain historical development of ejection seat

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of historical timeline slides; Instructor supervision required for Q&A; verification
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an ejection seat?
(b) Why was the ejection seat developed in the early 20th century?
(c) Can you name one early aircraft that utilized an ejection mechanism?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Early parachute systems in WWI • Development of the Martin-Baker seat • Introduction of the ACES II seat • Evolution of zero-zero capability • Integration of ballistic separation	L & Ds	PPT, IB	15 min
(b)	• Impact of high-speed jet requirements • Transition from manual to automatic triggers • Advancements in restraint systems • Development of canopy jettison mechanisms • Standardization of ejection protocols	L & Ds	PPT, PDF	20 min
(c)	• Modern safety features in ejection seats • Future trends in ejection technology • Comparison of international ejection systems • Role of simulation in seat testing • Maintenance and inspection history	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the requirements for ejection seats changed from propeller to jet aircraft.
(b) Describe the key difference between the Martin-Baker and ACES II ejection systems.
(c) Why is the zero-zero capability considered a critical requirement in modern ejection seats?

12. Home Assignment and Follow Up	Prepare a brief summary of the main parts of a typical ejection seat and their purposes to be discussed in the next lesson plan.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Main parts of a typical ejection seat and their purposes	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min